SFT and RL for Math Problem Using Qwen 2.5 Math 1.5B

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Abstract

This project explores the application of Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL) to improve the performance of the Qwen 2.5 Math 1.5B model on mathematical problem-solving tasks.

Starting from the Qwen 2.5 Math 1.5B model, which is highly sensitive to prompt formatting, I observed that a mismatch between the `r1_zero` prompt and the model's pretraining format led to a baseline accuracy of only 1.6% on the MATH dataset. Through supervised fine-tuning using reasoning traces from DeepSeek R1, performance improved substantially to 44%. I then applied reinforcement learning, specifically Group-Relative Policy Optimization (GRPO), to further optimize model reasoning under a reward-guided objective. This additional step raised the accuracy to 48%.

1 Introduction

Mathematical problem-solving with language models is a growing area of interest, particularly for its potential to enhance AI's utility in both educational and professional settings. Inspired by Jun Li's special talk in this class and the CS336 Assignment 5 [1], I chose this topic to pursue a more solid project compared to the agent-based work I did last year. Given the limited computing resources, achieving state-of-the-art performance through post-training in this project is challenging. Therefore, the primary goal is to gain hands-on experience in training language models to reason through math problems. Except of using HuggingFace Transformers to load the Qwen 2.5 Math 1.5B model and tokenizer and run forward passes, all code are built from stratch without using any of the training utilities (e.g., the `Trainer` class).

I combined Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL) to improve the reasoning capabilities of the Qwen 2.5 Math 1.5B model on the MATH dataset [2]. The Qwen Math series models are highly sensitive to prompt formatting. Due to a mismatch between the `r1_zero` prompt and the pretraining format of the model, the initial answer accuracy on the MATH dataset was only 1.6%. To address this, I first performed Supervised Fine-Tuning (SFT) using reasoning traces from DeepSeek R1 [3], which raised the accuracy to 44%. I then applied Group-Relative Policy Optimization (GRPO), a reward-guided RL method, to further improve performance to 48%.

Importantly, all training components—including loss computation, optimization, reward shaping, and batching—were implemented from scratch in PyTorch without reliance on open-source frameworks. This hands-on implementation offered both technical depth and empirical insight into the challenges of aligning LLMs with mathematical reasoning objectives.

2 Related Work

An exciting recent trend in language models is the use of chain-of-thought reasoning to improve performance across a variety of tasks.

Early chain-of-thought approaches finetuned language mod- els to solve simple mathematical tasks like arithmetic by using a "scratchpad" to break the problem into intermediate steps [4]. Other work prompts a strong model to “think step by step” be- fore answering, finding that this significantly improves performance on mathematical reasoning tasks like grade-school math questions [5].

Recent work has explored using more powerful reinforcement learning algorithms with verified rewards to improve reasoning performance. OpenAI’s o1 (and subsequent o3/o4) [6], and DeepSeek’s R1 [3] use policy gradient methods to train on math and code tasks where string matching or unit tests verify correctness, demonstrating remarkable improvements in competition math and coding performance. Later works such as Open-R1 [7], and SimpleRL-Zoo [8] confirm that pure reinforcement learning with verified rewards—even on models as small as 1.5B parameters—can improve reasoning performance.

3 Approach

3.1 SFT

As the goal is to improve the model’s reasoning ability, rather than finetune it to directly final correct answers, I finetune it to first generate a chain-of-thought reasoning trace from DeepSeek R1 [3] followed by an answer.

I follow the standard teacher forcing paradigm by minimizing the cross-entropy loss between the model’s predictions and the target reasoning trace. During training, only the tokens corresponding to the model’s response (not the prompt) are included in the loss computation. I implement `tokenize_prompt_and_output`, `compute_entropy`, `get_response_log_probs`, `masked_normalize`, `sft_microbatch_train_step` methods by myself.

Despite loading the model in bfloat16 and using FlashAttention-2, even an 80GB GPU does not have enough memory to support reasonable batch sizes. To use larger batch sizes, so I use gradient accumulation to simulate larger batch sizes. This improves training stability and model quality without requiring additional memory.

3.2 GRPO

In this project, I choose the Group-Relative Policy Optimization (GRPO) algorithm to improve the reasoning ability of the model. I implement a complete end-to-end training pipeline, including methods such as `compute_group_normalized_rewards`, `compute_grpo_clip_loss`, `masked_mean`, `grpo_microbatch_train_step`, and support for both on-policy and off-policy.

I set the `group_size` to 8 and enabled `use_std_normalization` to stabilize group-wise reward normalization in GRPO. Each rollout batch includes 256 samples, sampled with a temperature of 1.0 and a maximum length of 1024 tokens. I used a `grpo_clip` loss with a clipping range of 0.2, and accumulated gradients over 128 steps to simulate large-batch training.

4 Experiments

4.1 Model Selection

I selected the Qwen 2.5 Math 1.5B model for the task. This model is a modern, high-performance language model that supports Chain-of-Thought (CoT) reasoning, a crucial capability for solving complex mathematical problems. At the same time, this model does not require excessive computational resources. It can be fine-tuned on a single NVIDIA 80G H100 GPU. As the Azure $150 benefit credit cannot be used to access GPU resources, I personally cover the computational costs for this project.

In addition, I use the following prompt from the DeepSeek R1-Zero model [3]. RL with it shows clear accuracy improvements in a short number of steps. The following also included an experiment about question_only prompt for reality check.

```
A conversation between User and Assistant. The User asks a question, and the
Assistant solves it. The Assistant first thinks about the reasoning process
in the mind and then provides the User with the answer. The reasoning process
is enclosed within <think> </think> and answer is enclosed within
```

```
<answer> </answer> tags, respectively, i.e., <think> reasoning process
here </think> <answer> answer here </answer>.
User: {question}
Assistant: <think>
```

4.2 Benchmark

I use the MATH 12K dataset [2], which contains challenging high-school competition mathematics problems. The language model outputs will be evaluated by comparing them to a reference answer. I select around 5,000 problems from the dataset for validation, while the remaining 7,000 problems are used for training.

4.3 Evaluation method

The evaluation metric is clear—we test whether the model outputs exactly the correct answer. In math problems we assume that there is a known ground truth (e.g. `0.5`) but we cannot simply test whether the model outputs exactly 0.5—it can also answer `<answer> 1/2 </answer>`. Because of this, we must address the tricky problem of matching for semantically equivalent responses from the LLM when we evaluate MATH. I use a fast and fairly accurate answer parser used in recent work on reasoning RL [9]. This reward function is implemented at `alignment.drgrpo_grader.r1_zero_reward_fn`. This method both evaluate the format and answer corrnetness.

To evaluate language models, we have to generate continuations (responses) for a variety of prompts efficiently. This project use vLLM for offline batched inference. vLLM is a high-throughput and memory-efficient in- ference engine for language models that incorporates a variety of useful efficiency techniques (e.g., optimized CUDA kernels, PagedAttention for efficient attention KV caching, etc.).

4.4 Results

I evaluated the Qwen 2.5 Math 1.5B model on the MATH dataset under two different prompting strategies: `r1_zero` and `question_only`. The results are summarized in Table 1 and Table 2.

For the `r1_zero` prompt, the zero-shot model achieved only 10% format accuracy and 1.6% answer accuracy, indicating a severe mismatch between the prompt style and the model's pretraining distribution. After applying SFT, the format accuracy improved to 91%, and the answer accuracy increased significantly to 44%. Further applying GRPO led to a slight improvement in both format accuracy (93%) and answer accuracy (48%).

For the `question_only` prompt, the zero-shot performance was already relatively high, with an answer accuracy of 43%, suggesting that this prompt format aligns better with the model's pretraining. This accuracy match the result on the Qwen technical report [10]. GRPO further improved the answer accuracy to 59%.

These results demonstrate the effectiveness of SFT and RL in enhancing both structural fidelity and reasoning performance, particularly for more complex, structured prompt formats.

Setting	**Format Accuracy**	**Answer Accuracy**
Zero-shot	10%	1.6%
SFT	91%	44%
GRPO	93%	48%

Table 1: Experiment Result for r1_zero prompt

Setting	Answer Accuracy
Zero-shot	43%
GRPO	59%

Table 2: Experiment Result for question_only prompt

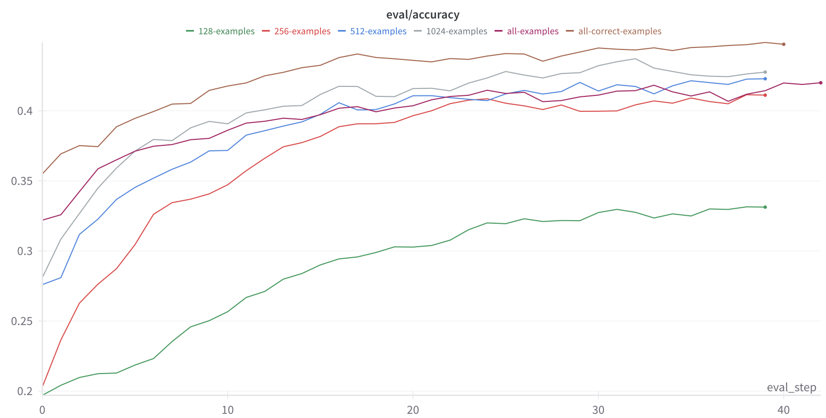

Figure 1: SFT Result

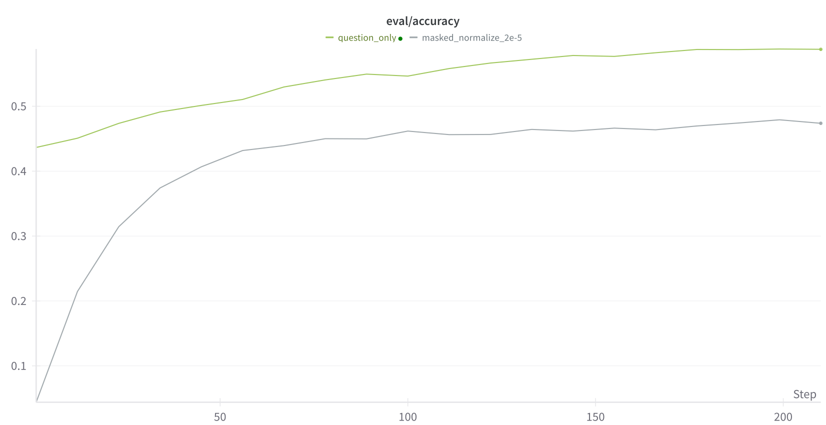

Figure 2: GRPO Result

5 Conclusion

In this project, I explored the use of Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL) to enhance the mathematical reasoning capabilities of the Qwen 2.5 Math 1.5B model. Despite limited computing resources, I was able to significantly improve performance on the MATH dataset—from a baseline accuracy of 1.6% to 44% after SFT, and further to 48% using Group-Relative Policy Optimization (GRPO). This project not only improved my technical understanding of LLM alignment, but also deepened my appreciation for the nuances of mathematical reasoning with large language models.

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